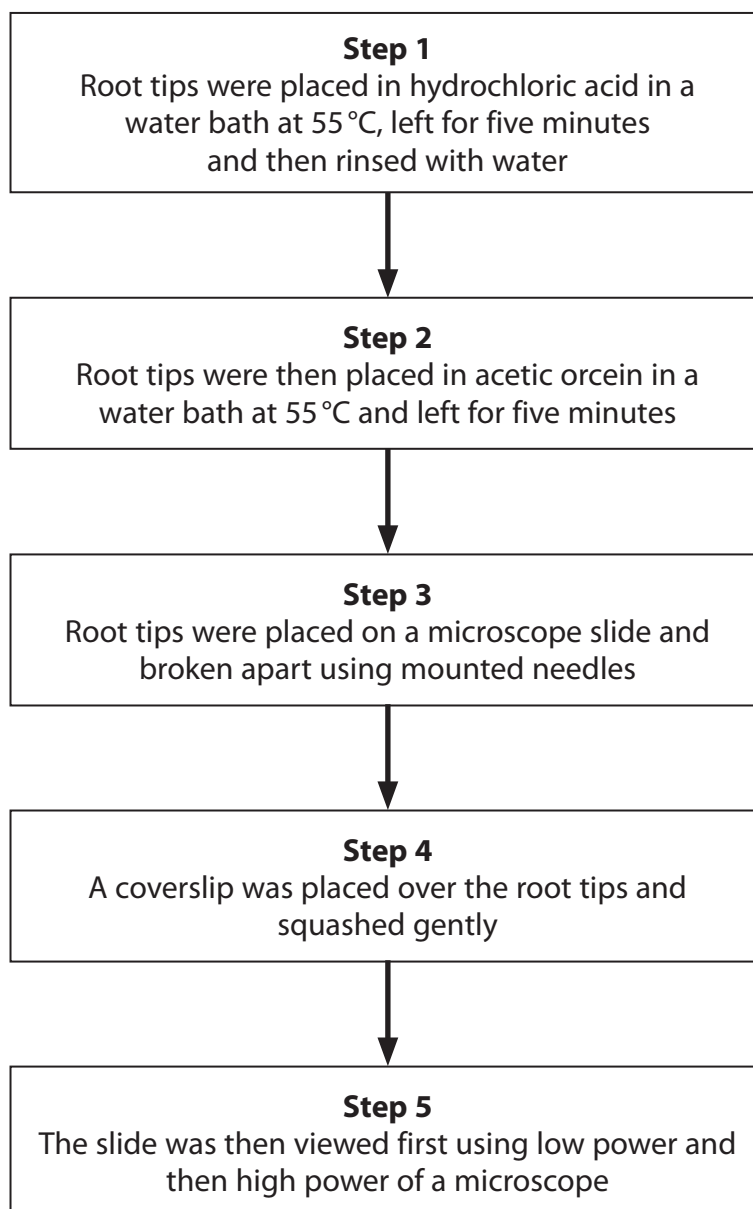


- 2 The flow chart shows how garlic root tips can be treated to observe mitosis in the root cells.



(a) (i) Give **one** reason why each of the following steps was taken.

(3)

Step 2

.....

.....

Step 4

.....

.....

Step 5

.....

.....

(ii) Explain **one** safety precaution that must be taken when carrying out this procedure.

(2)

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.....



(b) Describe how the procedure shown in the flow chart could be used to investigate the effect of pH on the mitotic index.

(5)

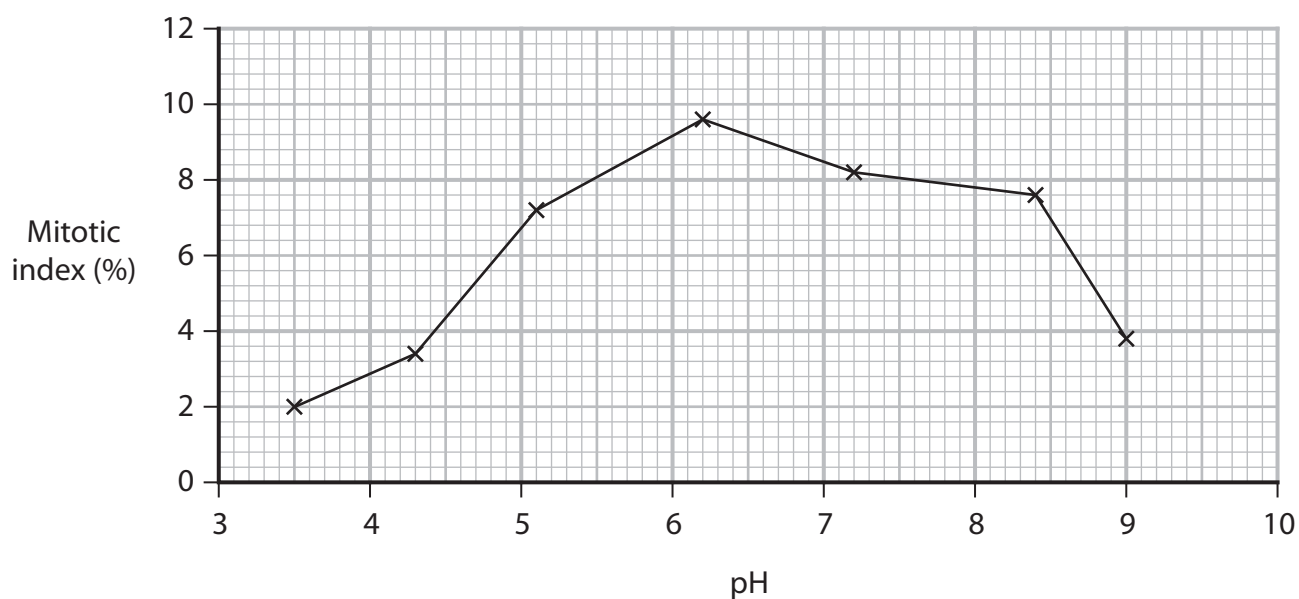


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(c) The graph shows the results of an investigation into the effect of pH on mitotic index.



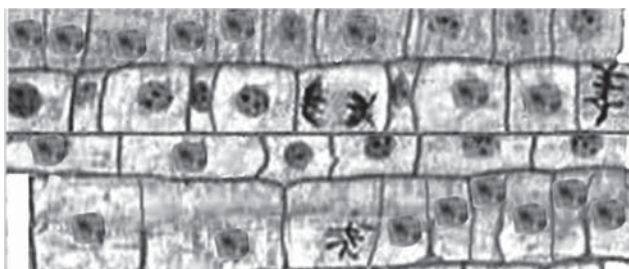
(i) Draw a suitable table for the results shown in the graph.

(3)



P 6 2 7 9 0 A 0 9 1 6

(ii) The photograph shows 35 cells from a root tip at one of the pH values investigated.



Calculate the mitotic index of this root tip.

Using the graph, determine the pH at which this root tip was grown.

(3)

Mitotic index%

pH

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(iii) Explain why the mitotic index is affected by pH, as shown by the graph.

(4)

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(Total for Question 2 = 20 marks)



P 6 2 7 9 0 A 0 1 1 1 6